EE1007
Introduction to Computer Science Laboratory

Lab 13
Bits, Characters, C Strings and structs

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Lab 13.1
Win the invoice lottery

- In this problem, you need to write a program which can find out all the invoices, which wins the lottery.
- The rule of winning lottery is that the invoices have to match the last 3, 4, 5, 6, 7, or all the numbers of the winning number.

頭獎	46626911 同期統一發票收執聯8位數號碼與頭獎號碼相同者獎金20萬元
二獎	同期統一發票收執聯末7 位數號碼與頭獎中獎號碼末7 位相同者各得獎金4萬元
三獎	同期統一發票收執聯末6 位數號碼與頭獎中獎號碼末6 位相同者各得獎金1萬元
四獎	同期統一發票收執聯末5 位數號碼與頭獎中獎號碼末5 位相同者各得獎金4千元
五獎	同期統一發票收執聯末4 位數號碼與頭獎中獎號碼末4 位相同者各得獎金1千元
六獎	同期統一發票收執聯末3 位數號碼與 頭獎中獎號碼末3 位相同者各得獎金2百元

Lab 13.1
Win the invoice lottery

- In the beginning, the program will take a sentence “The invoice lottery number is XXXXXXXX.” and a list of invoices user wants to check as inputs.
- Note that the range of the number of invoices is [1-10].
- Then the program will show the winning invoices, and how much money invoices won.
- In the end, the program should show the total price the invoices won.

# of matched number	Price
3	2 hundred
4	1 thousand
5	4 thousand
6	10 thousand
7	40 thousand
8	200 thousand

Lab 13.1
Win the invoice lottery

- The screen dialog should appear as follows:
  - First line shows the input sequence provided by the user, and the second line shows the input of a list of invoices user wants to check.
  - The rest of lines shows the output of the program.
- Example 1

```
The invoice lottery number is 12345678.
00000678 00005678 00045678

00000678 won 2 hundred dollars!
00005678 won 1 thousand dollars!
00045678 won 4 thousand dollars!
You totally won 5200 dollars !!!
```

Lab 13.1
Win the invoice lottery

◆The screen dialog should appear as follows:

- First line shows the input sequence provided by the user, and the second line shows the input of a list of invoices user wants to check.
- The rest of lines shows the output of the program.

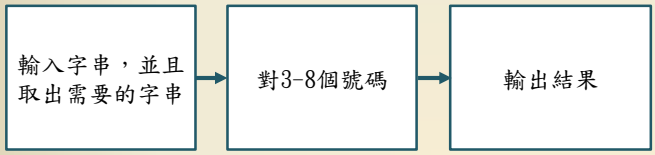
◆Example 2

```
The invoice lottery number is 12345678.
11111111 22222222 33333333

Won nothing. But, less is more.
```


Lab 13.1
Win the invoice lottery

◆思考方向:



Lab 13.1
Win the invoice lottery

- How to get invoice number:

```
//lottery_num = "The invoice lottery number is 12345678.";
lottery_num = lottery_num.substr(30, 8);
```

- How to check

```
for(int i=0; i<input_vec.size(); i++){
    if(input_vec[i] == lottery_num){
        cout << input_vec[i] << " won 200 thousand dollars!" << endl;
        num += 200000;
    }
    else if(input_vec[i].substr(1,7) == lottery_num.substr(1,7)){
        cout << input_vec[i] << " won 40 thousand dollars!" << endl;
        num += 40000;
    }
    else if(input_vec[i].substr(2,6) == lottery_num.substr(2,6)){
        cout << input_vec[i] << " won 10 thousand dollars!" << endl;
        num += 10000;
    }
    else if(input_vec[i].substr(3,5) == lottery_num.substr(3,5)){
        cout << input_vec[i] << " won 4 thousand dollars!" << endl;
        num += 4000;
    }
    else if(input_vec[i].substr(4,4) == lottery_num.substr(4,4)){
        cout << input_vec[i] << " won 1 thousand dollars!" << endl;
        num += 1000;
    }
    else if(input_vec[i].substr(5,3) == lottery_num.substr(5,3)){
        cout << input_vec[i] << " won 2 hundred dollars!" << endl;
        num += 200;
    }
}
```


Lab 13.1
Win the invoice lottery

- How to check:

```
for(int i=0; i<input_vec.size(); i++){
    if(input_vec[i] == lottery_num){
        cout << input_vec[i] << " won 200 thousand dollars!" << endl;
        num += 200000;
    }
    else if(input_vec[i].substr(1,7) == lottery_num.substr(1,7)){
        cout << input_vec[i] << " won 40 thousand dollars!" << endl;
        num += 40000;
    }
    else if(input_vec[i].substr(2,6) == lottery_num.substr(2,6)){
        cout << input_vec[i] << " won 10 thousand dollars!" << endl;
        num += 10000;
    }
    else if(input_vec[i].substr(3,5) == lottery_num.substr(3,5)){
        cout << input_vec[i] << " won 4 thousand dollars!" << endl;
        num += 4000;
    }
    else if(input_vec[i].substr(4,4) == lottery_num.substr(4,4)){
        cout << input_vec[i] << " won 1 thousand dollars!" << endl;
        num += 1000;
    }
    else if(input_vec[i].substr(5,3) == lottery_num.substr(5,3)){
        cout << input_vec[i] << " won 2 hundred dollars!" << endl;
        num += 200;
    }
}
```

Lab 13.2
Chances of blood type

- In this problem, you need to write a program that can find the chances of four kinds of blood types. The results depend on the blood types of the child's parents' blood type.
- There are four kinds of blood type: A, B, O and AB. One blood type is composed by 2 elements, which can be A, B or i.
- Please use the sample code to complete this problem.
- Your can't define any new variables. Please use the variables the sample code has.
- Moreover, you can only use cout or printf in the main function.
- You can only add new variable for loop control.

combinations	Blood type
AA, Ai	A
BB, Bi	B
AB	AB
ii	O

Lab 13.2
Chances of blood type

- The screen dialog should appear as follows:
  - First line and second lines show the input of the parents' gene provided by the user.
  - The rest of lines shows the chances of four kinds of blood types.
- Example 1

```
Ai
BB

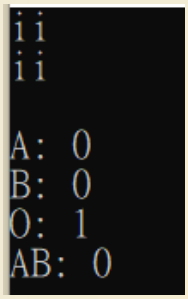
A: 0
B: 0.5
O: 0
AB: 0.5
```

Lab 13.2
Chances of blood type

◆The screen dialog should appear as follows:

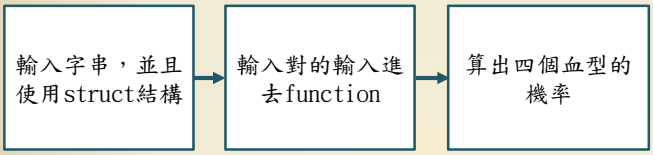
- First line and second lines show the input of the parents' gene provided by the user.
- The rest of lines shows the chances of four kinds of blood types.

◆Example 1



Lab 13.2
Chances of blood type

◆思考方向:



Lab 13.2
Chances of blood type

◆How to calculate chances of blood type :

```
if(MOM == 'A'){
    if(DAD == 'B')
        child->AB +=0.25;
    else
        child->A +=0.25;
}
else if(MOM == 'B'){
    if(DAD == 'A')
        child->AB +=0.25;
    else
        child->B +=0.25;
}
else if(MOM = 'i'){
    if(DAD == 'A')
        child->A +=0.25;
    else if(DAD == 'B')
        child->B +=0.25;
    else if(DAD == 'i')
        child->O +=0.25;
}
```


Lab 13.3
NCU Theater Reservation System(V)

- After create a membership, students want to have the latest information by the system of the NCU theater. Hence, the program needs members' phone number. Moreover, lots of people want to change the password and the phone number.
- Therefore, in this Lab, you are asked to update a new feature, which is the member's phone number and add a function which can change data of members from the old data.
- Once adding the new feature, the program needs to be modify the **member center function** and **load member data function**.
- First, at the moment of enrolling new membership in member center function, the program should ask user's phone number after the program ask the password of the member.

```
***NCU Theater Member Center***
Do you have the membership? (y/n) n
Would you like to sign up? (y/n) y
Enter Username: Bob
Enter Password: 1234
Enter Phone number: 0912345678
Welcome! You are now a member of NCU theater.
Please log in again.
```

Lab 13.3
NCU Theater Reservation System(V)

- Second, in the the **load member data function** the input data is modified as shown below.

3_1.txt Before

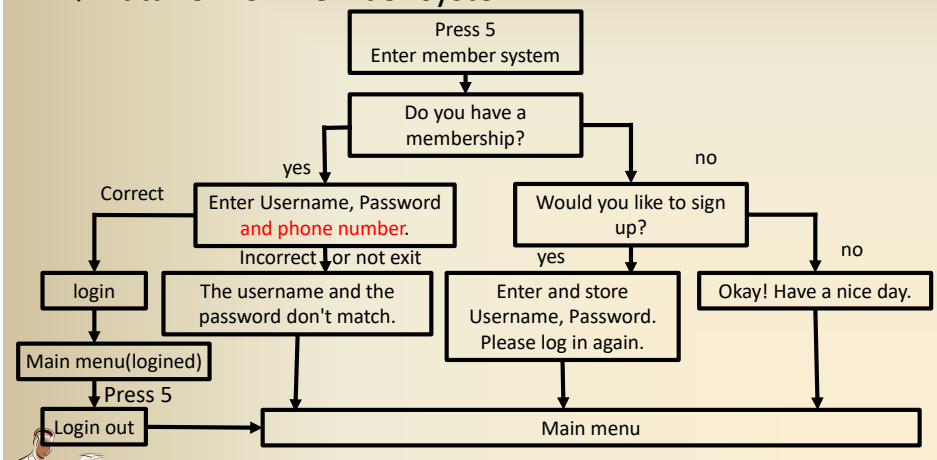
Name	Password
Andy	12345678
Kevin	227788
Jinx	456456
Akali	77777
Jax	878787

3_1.txt After

Name	Password	Phone
Andy	12345678	0975817468
Kevin	227788	0912345678
Jinx	456456	0812345678
Akali	77777	0812345678
Jax	878787	9032165498

Lab 13.3
NCU Theater Reservation System(V)

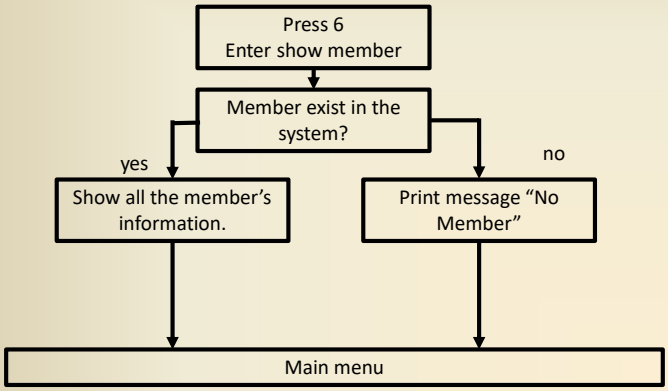
- Dataflow of member system



Lab 13.3

NCU Theater Reservation System(V)

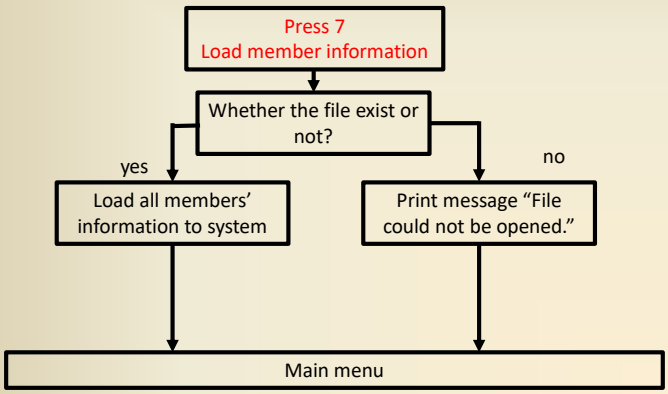
◆Dataflow of show member



Lab 13.3

NCU Theater Reservation System(V)

◆Dataflow of load member



Lab 13.3
NCU Theater Reservation System(V)

- Modification of member center function.

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to exit the system
Enter your selection Here: 5

***NCU Theater Member Center***
Do you have the membership? (y/n) n
Would you like to sign up? (y/n) y
Enter Username: Bob
Enter Password: 1234
Enter Phone number: 0912345678
Welcome! You are now a member of NCU theater.
Please log in again.

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to exit the system
Enter your selection Here:
```


Lab 13.3
NCU Theater Reservation System(V)

- After create a membership, students want to have the latest information by the system of the NCU theater. Hence, the program needs members' phone number. Moreover, lots of people want to change the password and the phone number.
- Therefore, in this Lab, you are asked to update a new feature, which is the member's phone number and add a function which can change data of members from the old data.
- Once a user enters 8 in main menu, the program can help user to change the member data.

Lab 13.3

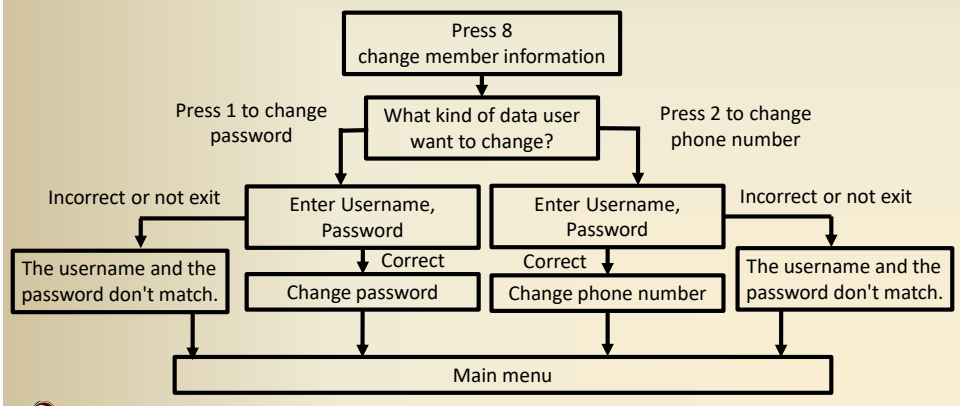
NCU Theater Reservation System(V)

- First, the program will print the message below.
  - "***Change member data***"
- Then, the program will ask what kind of data user wants to change.
- After user choose what kind of data user want to change, the program should ask user to enter username and the password to check whether the user have the authority.
- Note that the input of the phone number doesn't need to fool-proof.

Lab 13.3

NCU Theater Reservation System(V)

- Dataflow of change member data



Lab 13.3
NCU Theater Reservation System(V)

◆The screen dialog should appear as follows:

```
***NCU Theater Member Center***
Do you have the membership? (y/n) n
Would you like to sign up? (y/n) y
Enter Username: Bob
Enter Password: 1234
Enter Phone number: 0912345678
Welcome! You are now a member of NCU theater.
Please log in again.

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to exit the system
Enter your selection Here: 8

***Change member data***
Please type 1 to change password
Please type 2 to change phone number
Enter your selection Here: 1
Enter Username: Bob
Enter Password: 4321
The username and the password don't match.

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
```


Lab 13.3
NCU Theater Reservation System(V)

◆The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to exit the system
Enter your selection Here: 8

***Change member data***
Please type 1 to change password
Please type 2 to change phone number
Enter your selection Here: 1
Enter Username: Bob
Enter Password: 1234
Enter the new password: 0000

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
```

Lab 13.3
NCU Theater Reservation System(V)

◆The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to exit the system
Enter your selection Here: 8

***Change member data***
Please type 1 to change password
Please type 2 to change phone number
Enter your selection Here: 2
Enter Username: Bob
Enter Password: 0000
Enter the new phone number: 0911111111

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
```


Lab 13.3
NCU Theater Reservation System(V)

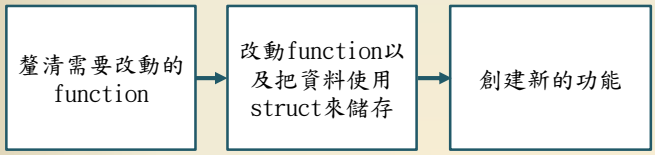
◆Requirement:

- If the user enters a wrong input, show the message **"Invalid input, please enter your selection again:"**

◆Notice that the program should work at normal mode when there is no one enter the member system.

Lab 13.3
NCU Theater Reservation System(V)

◆思考方向:



Lab 13.3
NCU Theater Reservation System(V)

◆How to change member data :

```
if (choice == 1) {

    cout << "Enter Username: " ;
    cin >> username;
    cout << "Enter Password: " ;
    cin >> password;

    for (int i = 0; i < 30; i++) {
        if (username == member_arr[i].name) {
            if (password == member_arr[i].password) {
                cout << "Enter the new password: ";
                cin >> password;
                member_arr[i].password = password;
                //cout << endl;
            } else if (password != member_arr[i].password) {
                cout << "The username and the password don't match." << endl << endl;
                return;
            }
        }
    }

    cout << endl;
}
```

Lab 13.3
NCU Theater Reservation System(V)

- How to change member data :

```
else if (choice == 2) {
    cout << "Enter Username: " ;
    cin >> username;
    cout << "Enter Password: " ;
    cin >> password;

    for (int i = 0; i < 30; i++) {
        if (username == member_arr[i].name) {
            if (password == member_arr[i].password) {
                cout << "Enter the new phone number: ";
                cin >> phone;
                member_arr[i].phone = phone;
                cout << endl;
            } else if (password != member_arr[i].password) {
                cout << "The username and the password don't match." << endl << endl;
                return;
            }
        }
    }
}
```


Lab 13.4
Rout assignment

- In a city there are n bus drivers. Also there are n morning bus routes and n afternoon bus routes with various lengths.
- Each driver is assigned one morning route and one evening route.
- For any driver, if his total route length for a day exceeds d, he has to be paid overtime. The rate of the overtime is r money / meter.
- Your task is to assign one morning route and one evening route to each bus driver so that the total overtime amount that the authority has to pay is minimized.

Lab 13.4
Rout assignment

- Input description:
  - The first line of each test case has three integers n, d and r, as described above.
  - In the second line, there are n space separated integers which are the lengths of the morning routes given in meters.
  - Similarly the third line has n space separated integers denoting the evening route lengths.
  - The lengths are positive integers less than or equal to 10000.
  - The end of input is denoted by a case with three 0's.
- For each test case, print the minimum possible overtime amount that the authority must pay.



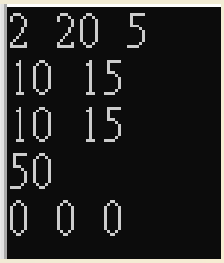
Lab 13.4
Rout assignment

- Please use the following struct to complete this problem. Otherwise, you will get 0 point.
  - Struct driver{
  - int distance;
  - int fine;
  - };
- The range of the three variable is shown below.
  - $1 \le n \le 100$
  - $1 \le d \le 10000$
  - $1 \le r \le 5$



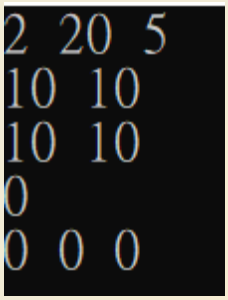
Lab 13.4
Rout assignment

- The screen dialog should appear as follows:
- Example 1:



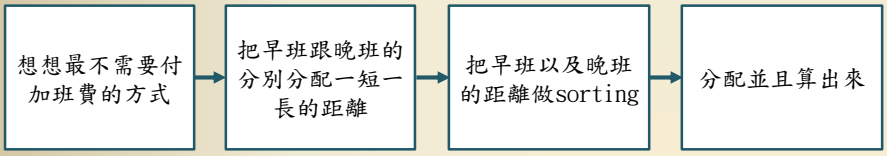
Lab 13.4
Rout assignment

- The screen dialog should appear as follows:
- Example 2:



Lab 13.4
Rout assignment

◆思考方向:



Lab 13.4
Rout assignment

◆How to calculate overtime :

再把sorting完的早班以及晚班的路徑分配並且計算加班費

```
int fine = 0;

for (int i = 0; i < n; ++i)
    if (morning[i] + night[n - i - 1] > d)
        fine += ((morning[i] + night[n - i - 1] - d) * r);
```



Evaluation



- Lab12.1(30%)
  - 5 testcase, 6% each.
- Lab12.2(30%)
  - 5 testcase, 6% each.
- Lab12.3(30%)
  - 5 testcase, 6% each.
- Lab12.4(20%)
  - 5 testcase, 4% each.

